

IHTC Model Transformations

Maksym Byelko

July 2026

MOVES / MODEL TRANSFORMATIONS

Decisions

D1: What day in the planning horizon will each patient be admitted?

D2: What operating theatre will each patient use?

D3: What recovery room will each patient use?

D4: What rooms will each nurse cover on their assigned shifts?

Model Transformations

All patient transformations operate only in the **PATIENTS** phase and maintain **RoomAvailability** across the complete stay interval. Nurse transformations operate only in the **NURSES** phase, after T17 confirms that every mandatory patient has an admission.

0.1 Patient Admission and Removal

- **T1.**

Type: atomic.

Description: Admit a feasible optional unscheduled patient to a day, room, and theatre.

Classes used: HospitalInstance, OptimisationState, Patient, Admission, SurgeonAvailability, OperatingTheatreAvailability, RoomAvailability

Purpose: to expand the schedule while maintaining full-stay room capacity and age-group compatibility.

- **T2.**

Type: atomic.

Description: Admit a feasible optional unscheduled patient with the minimum surgery duration.

Classes used: HospitalInstance, OptimisationState, Patient, Admission, SurgeonAvailability, OperatingTheatreAvailability, RoomAvailability

Purpose: to add a patient who is easier to fit into the schedule.

- **T3.**

Type: atomic.

Description: Admit an optional unscheduled patient using the earliest eligible surgeon availability.

Classes used: HospitalInstance, OptimisationState, Patient, Admission, Surgeon, SurgeonAvailability, OperatingTheatreAvailability, RoomAvailability

Purpose: to insert a patient in a way that is more strongly guided by actual resource availability, especially surgeon and theatre constraints.

- **T4.**

Type: atomic.

Description: Remove a scheduled optional patient and release their full room stay.

Classes used: HospitalInstance, OptimisationState, Patient, Admission, RoomAvailability, DeletedAdmissionsTracker

Purpose: to free up resources while only sacrificing an optional patient, which helps repair overloaded or poor-quality schedules.

- **T5.**
Type: atomic.
Description: Remove any scheduled patient and release their full room stay.
Classes used: HospitalInstance, OptimisationState, Patient, Admission, RoomAvailability, DeletedAdmissionsTracker
Purpose: to make a larger change to the schedule and free capacity, even if that means removing a more important patient.
- **T6.**
Type: joint (T4 + T1).
Description: Replace one optional scheduled patient with a different optional unscheduled patient.
Classes used: HospitalInstance, OptimisationState, Patient, Admission, RoomAvailability, SurgeonAvailability, OperatingTheatreAvailability, DeletedAdmissionsTracker
Purpose: to swap one optional patient for another, testing whether a different patient gives a better schedule without changing the total number of optional patients.
- **T7.**
Type: joint (T5 + T1).
Description: Replace any scheduled patient with a different optional unscheduled patient.
Classes used: HospitalInstance, OptimisationState, Patient, Admission, RoomAvailability, SurgeonAvailability, OperatingTheatreAvailability, DeletedAdmissionsTracker
Purpose: to replace part of the current schedule with a new optional patient, allowing a more disruptive change in the set of admitted patients.

0.2 Patient Reassignment

- **T8.**
Type: atomic.
Description: Move a scheduled patient to another compatible room on the same admission day.
Classes used: HospitalInstance, OptimisationState, Patient, Admission, Room, RoomAvailability
Purpose: to reassign a patient to a different feasible room in order to improve room usage or reduce incompatibilities.
- **T9.**
Type: atomic.
Description: Move a scheduled patient to a different feasible admission day while retaining room and theatre.
Classes used: HospitalInstance, OptimisationState, Patient, Admission, SurgeonAvailability, OperatingTheatreAvailability, RoomAvailability
Purpose: to move a patient to a different feasible day in order to improve schedule balance or resource usage.
- **T10.**
Type: atomic.
Description: Move a scheduled patient's operation to another available theatre on the same day.
Classes used: HospitalInstance, OptimisationState, Patient, Admission, OperatingTheatre, OperatingTheatreAvailability
Purpose: to reassign a patient to a different feasible theatre in order to improve theatre allocation.
- **T11.**
Type: joint (T8 + T9 + T10).
Description: Jointly change a patient's room, admission day, and theatre.
Classes used: HospitalInstance, OptimisationState, Patient, Admission, RoomAvailability, OperatingTheatreAvailability, SurgeonAvailability
Purpose: to make a broader coordinated change to one patient's allocation across several decision variables.
- **T12.**
Type: joint (T8 + T9).
Description: Jointly change a patient's room and admission day while retaining theatre.
Classes used: HospitalInstance, OptimisationState, Patient, Admission, RoomAvailability, SurgeonAvailability, OperatingTheatreAvailability
Purpose: to jointly adjust accommodation and timing in a way that may improve feasibility or quality more than changing one variable alone.
- **T13.**
Type: joint (T9 + T10).
Description: Jointly change a patient's admission day and theatre while retaining room.

Classes used: HospitalInstance, OptimisationState, Patient, Admission, OperatingTheatreAvailability, SurgeonAvailability, RoomAvailability

Purpose: to jointly modify the timing and surgical allocation of a patient while keeping the room unchanged.

0.3 Nurse Assignment

- **T14.**

Type: atomic.

Description: Assign an available, sufficiently skilled nurse to an active room-day-shift slot.

Classes used: HospitalInstance, OptimisationState, Nurse, NurseWorkingShift, HospitalisationShift, RoomShiftAssignment, Admission, PatientDayDemand

Purpose: to create cover only where current patient admissions generate nursing demand.

- **T15.**

Type: atomic.

Description: Remove one nurse-to-room-shift assignment.

Classes used: HospitalInstance, OptimisationState, Nurse, HospitalisationShift, RoomShiftAssignment

Purpose: to remove an existing nurse-to-room assignment and allow the optimisation to repair or rebalance staffing decisions.

- **T16.**

Type: joint (T15 + T14).

Description: Move a nurse to a different active room-day-shift in another room.

Classes used: HospitalInstance, OptimisationState, Nurse, NurseWorkingShift, HospitalisationShift, RoomShiftAssignment, Admission, PatientDayDemand

Purpose: to rebalance nursing cover while preserving the number of assignments.

- **T17.**

Type: atomic.

Description: Change the optimisation state from PATIENTS to NURSES once every mandatory patient is scheduled.

Classes used: HospitalInstance, OptimisationState, Patient

Purpose: to freeze patient scheduling and enable nurse assignment only after the mandatory admission threshold is reached.